

Vijayasree Garapati

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EDUCATION

University of California, Los Angeles (UCLA), Los Angeles, CA | Master of Engineering Expected, August, 2025

- Major in Artificial Intelligence - 3.95 GPA
- Courses: Neural Networks & Deep learning , Large Scale Machine learning, Natural Language Processing , Large Scale Data Mining, Large Scale Social and Complex networks, Management Communications , Business & Data Analytics , Entrepreneurship for Engineers

Birla Institute of Science & Technology, Dubai, UAE | Bachelor of Engineering 2022

- Major in Computer Science & Minor in Data Science - 9.55/10 CGPA
- Top 5% of class and placed in distinction division

TECHNICAL SKILLS

- Programming Languages: JavaScript | Typescript | Python | SQL | HTML & CSS | Java | C
- Web Development: React | Redux | Vue | Node | Nest Js | MongoDB | PostgreSQL | RabbitMQ | Git | Github | Jira | Jest | Redis | Cypress
- Machine Learning: Scikit-learn | PyTorch | Tensorflow | Pandas | Numpy | Matplotlib | Keras | Seaborn | Tableau

EXPERIENCE

Easygenerator Dubai, UAE

Full Stack Developer 08/2022 - 07/2024

- Architected and deployed backend services using Nest Js, PostgreSQL, MongoDB and TypeScript, supporting feature delivery across a cloud-native platform used by 2,000+ educators globally
- Improved frontend performance by 40% by implementing lazy loading with Intersection Observer and optimizing component rendering, enhancing UX across devices
- Built and scaled distributed systems for features like the LMS and Outline Builder, leveraging asynchronous task queues, API throttling, and modular service boundaries to ensure performance at high concurrency

Deriv Dubai, UAE

Front End Intern 01/2022 - 07/2022

- Contributed to cloud-hosted, real-time trading applications using React, WebSocket, and TypeScript, enhancing responsiveness and cross-session consistency for thousands of users
- Implemented service-layer logic in a microfrontend architecture, integrating external APIs, caching layers, and browser persistence for robust trading workflows
- Worked in Linux containerized environments with Git and Docker; participated in Agile-based iterations to troubleshoot deployment issues across compute services

Tendercare FZE Dubai, UAE

Software Engineering Intern 06/2020 - 08/2020

- Developed responsive web interfaces with Angular CLI, JavaScript, and HTML/CSS, revamping UI performance across desktop and mobile clients; participated in code debugging and performance profiling to identify inefficiencies in DOM rendering and event handling
- Executed targeted digital campaigns on LinkedIn and Twitter, collaborating with marketing team to revamp content strategy; efforts resulted in a 25% increase in engagement, strengthening company's online presence

SELECTED PROJECTS

University of California, Los Angeles Los Angeles, California

Student 09/2024 - Present

- Keystroke Prediction via sEMG (Deep Learning): Compared CNN+BiLSTM and Conformer models; reduced CER by over 50%, showcasing expertise in temporal modeling and signal processing
- LLM Hallucination Reduction (AI/ML/NLP): Built a token-level decoding pipeline for GPT-2 and LLaMA-3.2 using PyTorch and classifier-guided reranking; achieved 91% decoding accuracy with improved truthfulness
- Federated Learning with Privacy (Cloud, Big Data, Distributed Systems): Engineered adaptive noise-scaling for differentially private learning using Laplace/Gaussian distributions, overhauling model accuracy by 4% on MNIST under fixed ϵ -budgets

Birla Institute of Science & Technology Dubai, UAE

Student 08/2019 - 01/2022

- Nuclear Segmentation with Deep Learning (Medical Imaging, Deep Learning): developed and benchmarked U-Net, Grid Attention U-Net, and Capsule Networks for instance segmentation on MoNuSeg and MoNuSAC datasets; achieved Dice score of 0.86 and Jaccard index of 0.78, validating model accuracy through systematic evaluation pipelines
- Hybrid Recommendation Engine (Web + AI): Designed large-scale retrieval systems using TF-IDF, ResNet-50 embeddings, and Ball Tree KNN, enabling multimodal search on Amazon's Food dataset
- Fake News Detection (NLP, Big Data): Trained ensemble models including SVM, Voting Classifier, and MLPs, reaching 94% accuracy on labeled social data